

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 4.1 — Managing Compute Engine Resources

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Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 4: Ensuring successful operation of a cloud solution (~20 % of the exam)**. You must know how to remotely connect to instances, view VM inventory, work with snapshots, and work with images.

1. Remotely Connecting to Instances

SSH Connections (Linux VMs)

```
# Standard SSH via gcloud (recommended)
gcloud compute ssh my-vm --zone=us-central1-a

# SSH with a specific user
gcloud compute ssh alice@my-vm --zone=us-central1-a

# SSH through IAP tunnel (no external IP needed)
gcloud compute ssh my-vm --zone=us-central1-a --tunnel-through-iap

# SSH with port forwarding
gcloud compute ssh my-vm --zone=us-central1-a \
  --ssh-flag="-L 8080:localhost:80"

# SSH and run a command
gcloud compute ssh my-vm --zone=us-central1-a \
  --command="sudo systemctl status nginx"

# SCP files to/from a VM
gcloud compute scp local-file.txt my-vm:~/remote-file.txt --zone=us-central1-a
gcloud compute scp my-vm:~/remote-file.txt local-file.txt --zone=us-central1-a

# SCP a directory
gcloud compute scp --recurse my-vm:~/logs/ ./local-logs/ --zone=us-central1-a
```

RDP Connections (Windows VMs)

```
# Set/reset Windows password
gcloud compute reset-windows-password my-windows-vm \
  --zone=us-central1-a \
  --user=admin

# This returns: IP address, username, and password
# Use these with an RDP client (Remote Desktop Connection)
```

Serial Console Access

```
# Connect to serial console (troubleshooting — when SSH doesn't work)
gcloud compute connect-to-serial-port my-vm --zone=us-central1-a

# View serial port output (boot logs)
gcloud compute instances get-serial-port-output my-vm --zone=us-central1-a
```

Connection Methods Summary

Method	When to Use	External IP Required?
<code>gcloud compute ssh</code>	Standard SSH	Yes (or use IAP)
IAP tunnel	VM has no external IP	No
Serial console	SSH/RDP not working, boot issues	No
RDP	Windows VMs	Yes
Cloud Console SSH	Browser-based SSH	Yes (or use IAP)

2. Viewing Current Running VM Inventory

Listing Instances

```
# List all VMs in the project
gcloud compute instances list

# List VMs in a specific zone
gcloud compute instances list --filter="zone:us-central1-a"

# List only running VMs
gcloud compute instances list --filter="status=RUNNING"

# List VMs with specific labels
gcloud compute instances list --filter="labels.environment=production"

# Format output as table with specific columns
gcloud compute instances list \
  --format="table(name, zone, machineType.basename(), status, networkInterfaces[0].networkName)"

# Output as JSON
gcloud compute instances list --format=json

# Output as CSV
gcloud compute instances list --format=csv

# List VMs sorted by creation time
gcloud compute instances list --sort-by=creationTimestamp
```

Describing Instances

```
# Get detailed information about a VM
gcloud compute instances describe my-vm --zone=us-central1-a

# Get specific fields
gcloud compute instances describe my-vm \
  --zone=us-central1-a \
  --format="get(networkInterfaces[0].networkIP) "

# Get the external IP
gcloud compute instances describe my-vm \
  --zone=us-central1-a \
  --format="get(networkInterfaces[0].accessConfigs[0].natIP) "

# Get the machine type
gcloud compute instances describe my-vm \
  --zone=us-central1-a \
  --format="get(machineType) "
```

Instance Details (Key Properties)

Property	Description
Instance ID	Unique numeric identifier
Name	User-defined name
Zone	Zone where the VM is running
Machine Type	CPU/memory configuration
Status	RUNNING, STOPPED, TERMINATED, STAGING, etc.
Internal IP	Private IP address
External IP	Public IP (if assigned)
Disks	Boot disk and additional disks
Network Tags	Tags for firewall rules
Service Account	Attached service account
Labels	Key-value metadata for organization
Metadata	Custom key-value data (startup scripts, etc.)

Managing VM State

```
# Stop a VM (preserves disk, releases CPU/memory)
gcloud compute instances stop my-vm --zone=us-central1-a

# Start a stopped VM
gcloud compute instances start my-vm --zone=us-central1-a

# Reset (hard reboot) a VM
gcloud compute instances reset my-vm --zone=us-central1-a

# Suspend a VM (save memory state to disk)
gcloud compute instances suspend my-vm --zone=us-central1-a

# Resume a suspended VM
gcloud compute instances resume my-vm --zone=us-central1-a

# Delete a VM
gcloud compute instances delete my-vm --zone=us-central1-a

# Delete a VM but keep its boot disk
gcloud compute instances delete my-vm --zone=us-central1-a --keep-disks=boot
```

Modifying a VM

```
# Change machine type (VM must be stopped)
gcloud compute instances set-machine-type my-vm \
  --zone=us-central1-a \
  --machine-type=e2-standard-8

# Add/update labels
gcloud compute instances add-labels my-vm \
  --zone=us-central1-a \
  --labels=environment=staging,owner=alice

# Remove labels
gcloud compute instances remove-labels my-vm \
  --zone=us-central1-a \
  --labels=owner

# Add metadata
gcloud compute instances add-metadata my-vm \
  --zone=us-central1-a \
  --metadata=key1=value1

# Update tags
gcloud compute instances add-tags my-vm \
  --zone=us-central1-a \
  --tags=new-tag

# Move a VM to a different zone (within the same region)
gcloud compute instances move my-vm \
  --zone=us-central1-a \
  --destination-zone=us-central1-b
```

3. Working with Snapshots

What Are Snapshots?

- **Point-in-time copies** of persistent disk data
- Used for **backups**, **disaster recovery**, and **disk cloning**
- **Incremental** — Only changes since the last snapshot are stored (saves space and cost)
- **Global resource** — Can be used to create disks in any region
- Stored in Cloud Storage (managed by Google)

Creating Snapshots

```
# Create a snapshot of a disk
gcloud compute snapshots create my-snapshot \
  --source-disk=my-vm \
  --source-disk-zone=us-central1-a \
  --description="Backup before upgrade"

# Create a snapshot with labels
gcloud compute snapshots create my-snapshot \
  --source-disk=my-disk \
  --source-disk-zone=us-central1-a \
  --labels=purpose=backup,created-by=admin

# Create a snapshot of a specific disk (not boot disk)
gcloud compute snapshots create data-snapshot \
  --source-disk=my-data-disk \
  --source-disk-zone=us-central1-a

# Create a snapshot and store it in a specific location
gcloud compute snapshots create my-snapshot \
  --source-disk=my-disk \
  --source-disk-zone=us-central1-a \
  --storage-location=us
```

Viewing Snapshots

```
# List all snapshots
gcloud compute snapshots list

# Describe a snapshot
gcloud compute snapshots describe my-snapshot

# Filter snapshots by label
gcloud compute snapshots list --filter="labels.purpose=backup"

# List snapshots sorted by creation time
gcloud compute snapshots list --sort-by=creationTimestamp
```

Deleting Snapshots

```
# Delete a snapshot
gcloud compute snapshots delete my-snapshot

# Delete multiple snapshots
gcloud compute snapshots delete snapshot-1 snapshot-2 snapshot-3
```

Important: Deleting a snapshot does not affect the disk or any disks created from it. Google automatically manages the incremental chain — deleting a middle snapshot transfers the data to the next snapshot.

Scheduling Snapshots

Snapshot schedules automate regular backups:

```
# Create a snapshot schedule
gcloud compute resource-policies create snapshot-schedule my-schedule \
  --region=us-central1 \
  --max-retention-days=14 \
  --on-source-disk-delete=keep-auto-snapshots \
  --daily-schedule \
  --start-time=02:00 \
  --storage-location=us

# Create a weekly schedule
gcloud compute resource-policies create snapshot-schedule weekly-backup \
  --region=us-central1 \
  --max-retention-days=30 \
  --weekly-schedule-from-file=schedule.json \
  --storage-location=us

# Create an hourly schedule
gcloud compute resource-policies create snapshot-schedule hourly-backup \
  --region=us-central1 \
  --max-retention-days=7 \
  --hourly-schedule=4 \
  --start-time=00:00

# Attach schedule to a disk
gcloud compute disks add-resource-policies my-disk \
  --resource-policies=my-schedule \
  --zone=us-central1-a

# Remove schedule from a disk
gcloud compute disks remove-resource-policies my-disk \
  --resource-policies=my-schedule \
  --zone=us-central1-a

# List snapshot schedules
gcloud compute resource-policies list --filter="region:us-central1"
```


Creating a Disk from a Snapshot

```
# Create a new disk from a snapshot
gcloud compute disks create new-disk \
  --source-snapshot=my-snapshot \
  --zone=us-centrall1-a \
  --type=pd-ssd

# Create a disk in a different zone/region (cross-region restore)
gcloud compute disks create dr-disk \
  --source-snapshot=my-snapshot \
  --zone=europe-west1-b
```

4. Working with Images

What Are Images?

- **Boot disk templates** containing the operating system and software configuration
- Used to **create new VMs** or **new boot disks**
- Types: **Public images** (Google-provided) and **Custom images** (user-created)

Public Images

Image Family	Project	OS
debian-12	debian-cloud	Debian 12
ubuntu-2204-lts	ubuntu-os-cloud	Ubuntu 22.04 LTS
centos-stream-9	centos-cloud	CentOS Stream 9
rhel-9	rhel-cloud	Red Hat Enterprise Linux 9
windows-2022	windows-cloud	Windows Server 2022
cos-stable	cos-cloud	Container-Optimized OS

```
# List available public images
gcloud compute images list

# List images from a specific project
gcloud compute images list --project=debian-cloud

# Describe an image
gcloud compute images describe debian-12-bookworm-v20240312 --project=debian-cloud
```

Creating Custom Images

```
# Create an image from a VM's boot disk (VM should be stopped)
gcloud compute images create my-custom-image \
  --source-disk=my-vm \
  --source-disk-zone=us-central1-a \
  --description="Custom image with nginx and app"

# Create an image from a snapshot
gcloud compute images create my-image-from-snapshot \
  --source-snapshot=my-snapshot \
  --description="Image from backup snapshot"

# Create an image from a file in Cloud Storage
gcloud compute images create my-imported-image \
  --source-uri=gs://my-bucket/my-image.tar.gz

# Create an image with a specific family
gcloud compute images create my-app-v2 \
  --source-disk=my-vm \
  --source-disk-zone=us-central1-a \
  --family=my-app-family

# Deprecate an old image
gcloud compute images deprecate my-app-v1 \
  --state=DEPRECATED \
  --replacement=my-app-v2
```

Image Families

- A **family** is a group of related images (e.g., different versions of your app)
- When you reference a family, you always get the **latest non-deprecated** image
- Use families in instance templates for automatic image updates

```
# Create a VM using the latest image from a family
gcloud compute instances create my-vm \
  --image-family=my-app-family \
  --image-project=PROJECT_ID \
  --zone=us-central1-a
```

Viewing Images

```
# List custom images in your project
gcloud compute images list --no-standard-images

# List images in a specific family
gcloud compute images list --filter="family=my-app-family"

# Describe a custom image
gcloud compute images describe my-custom-image
```

Deleting Images

```
# Delete an image
gcloud compute images delete my-custom-image

# Delete multiple images
gcloud compute images delete old-image-1 old-image-2
```

Snapshots vs. Images

Feature	Snapshot	Image
Purpose	Backup and restore disks	Create new VMs/boot disks
Scope	Any persistent disk (boot or data)	Boot disk only
Incremental	Yes	No (full copy)
Cross-region	Yes	Yes
Families	No	Yes
Use case	Disaster recovery, backup	VM golden images, templates
Speed	Slower to create VM from	Faster to create VM from

8. Managed Instance Groups (MIGs)

- Group of identical VMs created from an instance template
- Stateless MIGs: auto-scaling, auto-healing, rolling updates (web frontends, batch)
- Stateful MIGs: preserve per-instance state (disks, metadata, IPs) for databases, stateful apps

```
# Create MIG
gcloud compute instance-groups managed create my-mig \
  --template=MY_TEMPLATE \
  --size=3 \
  --zone=us-central1-a

# Enable autoscaling
gcloud compute instance-groups managed set-autoscaling my-mig \
  --zone=us-central1-a \
  --max-num-replicas=10 \
  --min-num-replicas=2 \
  --target-cpu-utilization=0.7

# Set autohealing
gcloud compute instance-groups managed set-autohealing my-mig \
  --zone=us-central1-a \
  --health-check=MY_HEALTH_CHECK \
  --initial-delay=300

# Rolling update
gcloud compute instance-groups managed rolling-action start-update my-mig \
  --version=template=NEW_TEMPLATE \
  --zone=us-central1-a \
  --max-surge=1 \
  --max-unavailable=0

# List instances in MIG
gcloud compute instance-groups managed list-instances my-mig --zone=us-central1-a
```

- Regional MIG: distributes instances across zones in a region for HA

```
gcloud compute instance-groups managed create my-regional-mig --template=MY_TEMPLATE --size=3 --zone=us-central1-a
```

- **Exam tip:** MIGs + Load Balancer is the standard pattern for scalable, highly available stateless services

9. Startup and Shutdown Scripts

- Startup script: runs as root when VM boots (after OS loads)
- Shutdown script: runs when VM receives stop/preemption signal (30-second window)
- Pass via metadata:

```
# Inline
gcloud compute instances create my-vm \
  --metadata=startup-script='#!/bin/bash
apt-get update
apt-get install -y nginx'
# From file
gcloud compute instances create my-vm \
  --metadata-from-file=startup-script=/local/path/startup.sh
# From Cloud Storage
gcloud compute instances create my-vm \
  --metadata=startup-script-url=gs://my-bucket/startup.sh
# Update metadata on running instance
gcloud compute instances add-metadata my-vm --metadata-from-file=startup-script=/path/script.sh
```

- View startup script logs: `sudo journalctl -u google-startup-scripts.service`
- **Exam tip:** Startup scripts are idempotent by design; use for software installation, config application

Exam Practice Questions

1. You need to create a backup of a production VM's boot disk before performing an upgrade. What should you do?
 - Answer: Create a **snapshot** of the boot disk: `gcloud compute snapshots create pre-upgrade-backup --source-disk=my-vm --source-disk-zone=us-central1-a`.
2. Your team needs daily automated backups of a disk with 14-day retention. How should you set this up?
 - Answer: Create a **snapshot schedule** with `--daily-schedule`, `--max-retention-days=14`, then attach it to the disk with `gcloud compute disks add-resource-policies`.
3. You need to create identical VMs across multiple zones using a standardized OS with your application pre-installed. What approach should you use?
 - Answer: Create a **custom image** from a configured VM (with your app installed), assign it to an **image family**, then use the image family in an **instance template** for a regional MIG.
4. How can you connect to a VM that is not booting properly (SSH is not working)?
 - Answer: Use the **serial console**: `gcloud compute connect-to-serial-port my-vm --zone=us-central1-a`. This lets you see boot logs and interact with the console.
5. You need to change a VM's machine type from e2-standard-4 to e2-standard-8. Can you do this while the VM is running?

- Answer: **No**. You must stop the VM first, change the machine type, then start it again:
`gcloud compute instances stop` , `gcloud compute instances set-machine-type` , `gcloud compute instances start` .

6. You deleted a snapshot that was part of an incremental chain. Will other snapshots be affected?

- Answer: **No**. Google automatically manages incremental snapshot data. When a snapshot is deleted, any data needed by later snapshots is transferred to them automatically.

Glossary

Auto-healing — A MIG feature that automatically detects unhealthy VM instances using a health check and replaces them without manual intervention.

Autoscaling — The automatic adjustment of the number of VM instances in a MIG based on load metrics such as CPU utilization, to match demand while controlling cost.

Boot Disk — The primary persistent disk attached to a Compute Engine VM that contains the operating system; used as the source for creating custom images.

CentOS Stream — A community-supported Linux distribution available as a public Compute Engine image; a continuously delivered upstream branch of RHEL.

Cloud Storage — GCP's object storage service; snapshots are stored by Google in Cloud Storage, and startup scripts can be stored and referenced from Cloud Storage buckets.

Compute Engine — GCP's Infrastructure-as-a-Service (IaaS) offering for creating and managing virtual machine instances.

Container-Optimized OS (cos-stable) — A Google-maintained Linux OS image optimized for running Docker containers on Compute Engine VMs.

CPU Utilization — The percentage of CPU capacity being used by a VM; used as a scaling metric for MIG autoscaling (e.g., `--target-cpu-utilization=0.7`).

Cross-Region Restore — The ability to create a disk in a different region from a snapshot, enabling disaster recovery in an alternate geographic location.

Custom Image — A user-created Compute Engine boot disk image containing a preconfigured operating system and software stack, used as a template for creating new VMs.

Debian — A popular open-source Linux distribution available as a public Compute Engine image (e.g., `debian-12` in the `debian-cloud` project).

Disaster Recovery (DR) — A set of policies, tools, and procedures for recovering infrastructure after a failure; snapshots are a key tool for DR on Compute Engine.

External IP — A publicly routable IP address assigned to a VM instance, enabling direct communication with the internet.

GCP (Google Cloud Platform) — Google's suite of cloud computing services, including Compute Engine and related infrastructure services.

GKE (Google Kubernetes Engine) — GCP's managed Kubernetes service; referenced as a consumer of Compute Engine VM infrastructure.

Golden Image — A custom VM image that has been configured, hardened, and validated as the standard template for deploying new VM instances across an environment.

HA (High Availability) — A design goal achieved by distributing VM instances across multiple zones using regional MIGs, ensuring continued operation if one zone fails.

Health Check — A GCP resource that periodically probes a VM or backend to determine if it is healthy; used by MIGs for auto-healing.

IAP (Identity-Aware Proxy) — A GCP service that enables SSH/RDP access to VMs over an encrypted tunnel without requiring a public external IP address (via `--tunnel-through-iap`).

ICMP (Internet Control Message Protocol) — A network diagnostic protocol; used in startup scripts and firewall rule context to test VM connectivity.

Image Family — A named group of related Compute Engine images (e.g., different versions of a custom application image) where referencing the family always returns the latest non-deprecated image.

Incremental Snapshot — A snapshot that stores only the data changed since the previous snapshot, reducing storage consumption and cost; all GCP snapshots after the first are incremental.

Instance ID — A unique numeric identifier automatically assigned to each Compute Engine VM instance by GCP.

Instance Template — A resource that defines the configuration (machine type, image, disk, network, metadata) for VM instances created by a Managed Instance Group.

Internal IP — A private IP address assigned to a VM within its VPC subnet, used for communication within the VPC network.

Label — A key-value metadata tag applied to GCP resources (including VMs and snapshots) used for organization, filtering, cost attribution, and automation.

Machine Type — The hardware configuration of a Compute Engine VM, specifying the number of vCPUs and amount of memory (e.g., `e2-standard-4`, `n2-standard-8`).

Metadata — Key-value pairs attached to a Compute Engine VM instance; used to pass startup scripts, configuration data, and custom information accessible from within the VM.

MIG (Managed Instance Group) — A group of identical Compute Engine VM instances created from an instance template, supporting autoscaling, auto-healing, and rolling updates.

Network Tag — A string label applied to a VM instance to make it a target for specific firewall rules or static routes.

OS Login — A GCP feature that allows SSH access to VMs using Google identities (IAM accounts) instead of manually managed SSH keys; enabled via the `enable-oslogin` metadata key.

pd-ssd — The Compute Engine persistent disk type backed by solid-state drives, offering lower latency and higher IOPS than standard persistent disks.

Persistent Disk — GCP's durable block storage for Compute Engine VMs that persists independently of the VM's lifecycle; the source for snapshots.

Preemption — The termination of a Spot VM by GCP when compute capacity is needed; a 30-second shutdown signal is sent, allowing shutdown scripts to run.

Public Image — A Compute Engine boot disk image provided and maintained by Google or third-party vendors (e.g., Debian, Ubuntu, Windows Server), available to all GCP projects.

RDP (Remote Desktop Protocol) — A Microsoft protocol for remote graphical desktop access; used to connect to Windows Compute Engine VMs.

Red Hat Enterprise Linux (RHEL) — A commercially supported Linux distribution available as a public Compute Engine image.

Regional MIG — A Managed Instance Group that distributes VM instances across multiple zones within a region for higher availability.

Resource Policy — A GCP resource that defines scheduled operations, such as snapshot schedules, and can be attached to disks or VMs.

Rolling Update — A MIG feature that gradually replaces instances running an old template version with instances running a new version, with configurable `--max-surge` and `--max-unavailable` settings.

SCP (Secure Copy Protocol) — A protocol for securely transferring files between a local machine and a remote server; used via `gcloud compute scp` to move files to/from VMs.

Serial Console — A low-level text interface to a VM that is accessible even when the operating system has not fully booted; used for troubleshooting boot failures and network issues.

Service Account — A GCP identity assigned to a VM instance to allow it to authenticate API calls to GCP services without requiring user credentials.

Shutdown Script — A script executed on a Compute Engine VM when it receives a stop or preemption signal, with a 30-second execution window to perform cleanup tasks.

Snapshot — A point-in-time copy of a Compute Engine persistent disk, stored by Google in Cloud Storage; used for backups, disaster recovery, and disk cloning.

Snapshot Schedule — An automated policy (resource policy) attached to a disk that triggers periodic snapshot creation and manages retention based on configured parameters.

Spot VM — A Compute Engine VM type that uses spare GCP capacity at a lower cost but can be reclaimed by GCP with 30 seconds notice when capacity is needed.

SSH (Secure Shell) — A cryptographic protocol for secure remote command-line access to Linux VMs; the primary connection method for Linux Compute Engine instances.

Startup Script — A script that runs automatically as root when a Compute Engine VM boots; used to install software, apply configuration, and perform initialization tasks.

Stateful MIG — A Managed Instance Group configured to preserve per-instance state (specific disks, metadata, and IP addresses) across instance recreation, suitable for stateful applications.

Stateless MIG — A Managed Instance Group where all instances are interchangeable and no per-instance state is preserved; suitable for web frontends and batch processing.

Suspend — A Compute Engine VM state that saves the in-memory state to disk and stops billing for CPU/memory, allowing the VM to be resumed later with its memory state intact.

Ubuntu — A popular open-source Linux distribution available as a public Compute Engine image (e.g., `ubuntu-2204-lts`).

VM (Virtual Machine) — A software-emulated computer running on physical hardware; the primary compute resource in Compute Engine.

Windows Server — Microsoft's server operating system available as a public Compute Engine image; requires RDP for remote access.

Zone — A geographically isolated deployment area within a GCP region; Compute Engine VM instances and disks are zonal resources.